Fidelity- and Uncertainty-Aware Network-Digital-Twin-Assisted Proactive Traffic Engineering in SDN

A. Proposed Research Problem

We consider a Software-Defined Network in which routing decisions must be made under rapidly varying traffic demands, link congestion, queue build-up, packet loss, and occasional capacity degradation. Conventional SDN traffic-engineering schemes typically rely on the most recently observed network state. As a result, they usually trigger routing actions only after congestion or performance degradation has already become visible.

A Network Digital Twin (NDT) provides an opportunity to move from such reactive control toward predictive network management. Existing NDT research has demonstrated the usefulness of virtual network representations for traffic prediction, simulation, optimization, and closed-loop network control. More recent systems have also coupled Digital Twins with reinforcement learning or SDN-based proactive control.

However, a fundamental issue remains: **future network states predicted by a Digital Twin are uncertain, and the Digital Twin itself may temporarily deviate from the physical network because of telemetry delays, incomplete measurements, model errors, or state synchronization latency**. Practical NDT studies explicitly identify synchronization between the physical and virtual networks as a central operational challenge.

Therefore, a controller that blindly selects the route appearing optimal under a single predicted future state can cause premature rerouting, route oscillations, incorrect preemption, and even worse congestion when the prediction is inaccurate.

We propose a **Fidelity- and Uncertainty-Aware Digital-Twin-Assisted Predictive Traffic Engineering framework**, in which the SDN controller does not optimize against a single forecast. Instead, it:

1. maintains a continuously synchronized Network Digital Twin;
2. measures the fidelity of the twin with respect to the physical network;
3. predicts a distribution or calibrated uncertainty region for future network states;
4. executes counterfactual simulations of alternative routing decisions inside the twin;
5. estimates the tail risk of congestion and SLA violations under each candidate decision;
6. explicitly accounts for the cost of changing an existing route; and
7. proactively modifies the physical network only when the predicted benefit is sufficiently robust to uncertainty.

The resulting problem is therefore fundamentally different from instantaneous shortest-path routing and from simple prediction-assisted routing.

B. Network Model

We represent the physical SDN network as a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}),$$

where $\mathcal{V}$ denotes the set of SDN switches and $\mathcal{E}$ denotes the set of directed communication links.

For each link $e \in \mathcal{E}$, let

$$C_e(t)$$

denote its available transmission capacity at control epoch t.

The controller observes a telemetry vector

$$\mathbf{s}_e(t) = [u_e(t), q_e(t), d_e(t), \ell_e(t), C_e(t)],$$

where

- $u_e(t)$ is link utilization,
- $q_e(t)$ is queue occupancy,
- $d_e(t)$ is measured delay,
- $\ell_e(t)$ is packet-loss probability, and
- $C_e(t)$ is currently available capacity.

Let

$$\mathcal{F}(t) = \{1, \ldots, F\}$$

denote the active flows.

Each flow f is characterized by

$$f = (s_f, d_f, \lambda_f, D_f^{\max}, L_f^{\max}),$$

where s_f and d_f denote the source and destination, λ_f denotes the traffic demand, and $D_f^{\max}$ and $L_f^{\max}$ denote the flow's maximum admissible delay and packet-loss probability.

For each flow f, the controller computes a set of feasible candidate paths

$$\mathcal{P}_f.$$

Define

$$A_{ep} = \begin{cases} 1, & e \in p, \\ 0, & \text{otherwise.} \end{cases}$$

C. Network Digital Twin

The Network Digital Twin maintains a virtual state

$$\widetilde{\mathbf{s}}_e(t)$$

for every physical link.

Unlike an ordinary network emulator, the twin continuously updates itself using telemetry from the physical SDN.

We define the instantaneous twin discrepancy as

$$\delta_e(t) = \sum_{k=1}^{K} \omega_k \frac{|s_{e,k}(t) - \widetilde{s}_{e,k}(t)|}{R_k + \epsilon},$$

where R_k normalizes telemetry dimension k,

$$\omega_k \geq 0, \qquad \sum_k \omega_k = 1.$$

We then define the **Digital-Twin fidelity score**

$$\phi_e(t) = \exp[-\kappa \delta_e(t)],$$

where

$$0 < \phi_e(t) \leq 1.$$

A value close to one indicates that the twin closely represents the physical network, whereas a small value indicates significant disagreement.

This variable is important because the proposed controller must distinguish between:

uncertain future network behavior

and

an inaccurate representation of the current physical network.

The latter issue is largely ignored when prediction outputs are treated as ground truth.

D. Multi-Step Future-State Prediction

At control epoch t, the twin predicts network evolution over a look-ahead horizon

$$\mathcal{H} = \{1, \ldots, H\}.$$

Rather than generating only a deterministic prediction such as

$$\widehat{\lambda}_f(t+h),$$

the proposed predictor generates a probability distribution, prediction interval, or scenario ensemble.

Let

$$\Omega = \{1, \ldots, S\}$$

denote the future scenarios generated by the NDT, with scenario probability

$$\pi_\omega, \qquad \sum_{\omega \in \Omega} \pi_\omega = 1.$$

Scenario ω specifies

$$\xi^{\omega}_{t+h} = \left\{ \widehat{\lambda}^{\omega}_{f,h}, \widehat{C}^{\omega}_{e,h}, \widehat{\ell}^{\omega}_{e,h}, \ldots \right\}.$$

The predictor may, for example, generate prediction intervals using probabilistic forecasting or conformal calibration. Recent work demonstrates that conformal methods can provide calibrated uncertainty information for traffic-demand prediction rather than only point forecasts.

Fidelity-dependent uncertainty

A key element of the proposed model is that forecast uncertainty is enlarged whenever the twin is poorly synchronized with the physical network.

Let $r^0_{e,h}$ denote the nominal uncertainty radius produced by the predictor. We define

$$r_{e,h} = r^0_{e,h} \left[1 + \beta(1 - \phi_e(t))\right], \qquad \beta \geq 0.$$

Thus,

$$\phi_e(t) \to 1 \quad \Rightarrow \quad r_{e,h} \to r^0_{e,h},$$

whereas decreasing twin fidelity increases the uncertainty region used by the routing optimizer.

The routing decision therefore automatically becomes more conservative when the Digital Twin is unreliable.

E. Predictive Routing Variables

For flow f, candidate path $p \in \mathcal{P}_f$, and future epoch h,

$$x_{fph} \in \{0, 1\}$$

indicates whether path p is assigned to flow f.

Hence,

$$\sum_{p \in \mathcal{P}_f} x_{fph} = 1, \qquad \forall f, h. \tag{C1}$$

Let

$$z_{fh} \in \{0, 1\}$$

indicate whether the route of flow f changes between two consecutive routing epochs.

F. Counterfactual Link Utilization

For future scenario ω, the predicted load on link e is

$$Y^{\omega}_{e,h} = \sum_{f \in \mathcal{F}} \sum_{p \in \mathcal{P}_f} A_{ep} \widehat{\lambda}^{\omega}_{f,h} x_{fph}.$$

The corresponding utilization is

$$\rho^{\omega}_{e,h} = \frac{Y^{\omega}_{e,h}}{\widehat{C}^{\omega}_{e,h}}. \tag{1}$$

Unlike conventional forecasting followed by shortest-path calculation, Eq. (1) represents a **counterfactual prediction**.

The future utilization depends not only on the predicted exogenous traffic but also on the routing decision being evaluated.

Thus, the twin evaluates the question

> "What will the future network state become if this particular routing configuration is installed?"

rather than simply asking

> "What will the network state be?"

This distinction is important because rerouting traffic itself changes future congestion.

G. Delay Model

Let the twin estimate link delay as

$$D^{\omega}_{e,h} = D^{\text{prop}}_{e} + g_e\left(\rho^{\omega}_{e,h}, q^{\omega}_{e,h}\right),$$

where $g_e(\cdot)$ is a monotonic queuing-delay model learned or calibrated from network telemetry.

For an analytical implementation, we can adopt an M/M/1-inspired approximation:

$$D^{\omega}_{e,h} = D^{\text{prop}}_{e} + \frac{1}{\mu_e - Y^{\omega}_{e,h}},$$

provided

$$Y^{\omega}_{e,h} < \mu_e.$$

The end-to-end delay experienced by flow f is therefore

$$D^{\omega}_{f,h} = \sum_{p \in \mathcal{P}_f} x_{fph} \sum_{e \in \mathcal{E}} A_{ep} D^{\omega}_{e,h}. \tag{2}$$

H. Packet-Loss Model

Let

$$p_{e,h}^{\omega} = g_e^L(\rho_{e,h}^{\omega}, q_{e,h}^{\omega})$$

denote the packet-loss probability estimated by the NDT.

Assuming conditionally independent link losses, the path success probability is

$$P_{f,h}^{\mathrm{succ},\omega} = \prod_{e \in p_f} (1 - p_{e,h}^{\omega}),$$

and therefore

$$L_{f,h}^{\omega} = 1 - P_{f,h}^{\mathrm{succ},\omega}. \tag{3}$$

For numerical optimization, we can alternatively use an additive reliability representation:

$$R_{f,h}^{\omega} = -\sum_{p} x_{fph} \sum_{e} A_{ep} \log(1 - p_{e,h}^{\omega}).$$

I. Congestion Risk

The controller should not change a route simply because its *mean predicted utilization* is slightly lower.

Instead, we define the congestion violation

$$V_{e,h}^{\omega} = \left[\rho_{e,h}^{\omega} - \rho_e^{\mathrm{safe}}\right]^+,$$

where

$$[x]^+ = \max(x, 0)$$

and typically

$$\rho_e^{\mathrm{safe}} < 1.$$

For example, we may use $\rho_e^{\mathrm{safe}} = 0.8$ to reserve headroom against bursts.

The proposed formulation controls not only expected congestion but also extreme predicted scenarios through Conditional Value-at-Risk:

$$\mathrm{CVaR}_{\alpha}(V_e).$$

Using the standard auxiliary representation,

$$\mathrm{CVaR}_{\alpha}(V) = \eta + \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} [V^{\omega} - \eta]^+. \tag{4}$$

For $\alpha = 0.95$, Eq. (4) emphasizes the worst 5% of predicted network conditions.

This is particularly important for proactive routing because a route whose *average* future utilization appears satisfactory may still have an unacceptable congestion tail.

J. SLA-Violation Risk

Define the delay violation

$$V_{f,h}^{D,\omega} = \left[D_{f,h}^{\omega} - D_f^{\max}\right]^+$$

and packet-loss violation

$$V_{f,h}^{L,\omega} = \left[L_{f,h}^{\omega} - L_f^{\max}\right]^+.$$

The controller can enforce risk constraints such as

$$\mathrm{CVaR}_{\alpha_D}\left(V_{f,h}^{D}\right) \le \epsilon_f^{D}, \qquad \forall f, h, \tag{C2}$$

and

$$\mathrm{CVaR}_{\alpha_L}\left(V_{f,h}^{L}\right) \le \epsilon_f^{L}. \tag{C3}$$

Hence, the controller does not regard the route as safe simply because its expected latency satisfies the SLA.

K. Route-Reconfiguration Cost

Aggressive proactive routing can create another problem: route oscillation.

Suppose the currently installed route is

$$x_{fp,h-1}.$$

We introduce

$$z_{fh} \ge x_{fph} - x_{fp,h-1}, \qquad \forall p. \tag{C4}$$

Since only one path is active, $z_{fh} = 1$ whenever flow f changes route.

A controller-level reconfiguration budget is imposed:

$$\sum_{f} z_{fh} \le B_h^{\mathrm{route}}, \qquad \forall h. \tag{C5}$$

This models finite SDN rule-installation capacity and prevents simultaneous rerouting of an excessive number of flows.

We may additionally impose a minimum route dwell time T_f^{dwell}:

$$\sum_{\tau=h}^{h+T_f^{\mathrm{dwell}}-1} z_{f\tau} \le 1. \tag{C6}$$

This constraint explicitly suppresses ping-pong routing.

L. Multi-Objective Optimization

The routing objective combines five quantities:

1. expected end-to-end latency;
2. packet-loss exposure;
3. network-load imbalance;
4. tail risk of future congestion/SLA violation; and
5. route-reconfiguration cost.

Define the normalized average delay

$$J_D = \frac{1}{H|\mathcal{F}|} \sum_{h,f,\omega} \pi_\omega \frac{D_{f,h}^{\omega}}{D_f^{\max}}.$$

Define the normalized loss cost

$$J_L = \frac{1}{H|\mathcal{F}|} \sum_{h,f,\omega} \pi_\omega \frac{L^\omega_{f,h}}{L^{\max}_f + \epsilon}.$$

Introduce

$$U^\omega_h \geq \rho^\omega_{e,h}, \qquad \forall e,$$

so that

$$J_U = \frac{1}{H} \sum_{h,\omega} \pi_\omega U^\omega_h$$

penalizes peak utilization.

Let

$$J_{\text{risk}} = \text{CVaR}_\alpha \left[\sum_h \gamma^{h-1} \left(\sum_e V^\omega_{e,h} + \sum_f V^{D,\omega}_{f,h} + \sum_f V^{L,\omega}_{f,h} \right) \right].$$

Finally,

$$J_R = \frac{1}{H} \sum_{h,f} c^R_f z_{fh}$$

represents SDN reconfiguration overhead.

We formulate the predictive routing problem as

$$\boxed{\begin{array}{lll} \textbf{P1}: & \min\limits_{\mathbf{x},\mathbf{z}} & w_D J_D + w_L J_L + w_U J_U + w_R J_{\text{risk}} + w_C J_R \\ & \text{s.t.} & \text{(C1)–(C6)}, \\ & & x_{fph} \in \{0,1\}, \\ & & z_{fh} \in \{0,1\}. \end{array}} \tag{P1}$$

where

$$w_D + w_L + w_U + w_R + w_C = 1.$$

For publication-quality experiments, we should not choose the weights only to produce favorable results. We should either derive them from SLA priorities or vary them systematically to generate Pareto trade-offs.

M. Proactive Nature of the Formulation

Suppose link e_1 currently operates at

$$\rho_{e_1}(t) = 0.50.$$

A conventional congestion-triggered controller would retain the route because the link is currently uncongested.

Assume, however, that the Digital Twin predicts

$$\rho_{e_1}(t+3) = 0.86$$

with a calibrated interval

$$[0.78, 0.96].$$

For an alternative route through e_2, counterfactual simulation predicts

$$\rho_{e_2}(t+3) = 0.63,$$

with interval

$$[0.58, 0.69].$$

Problem **P**1 may select the second route at the present control epoch even though neither link is currently congested.

The decision is therefore driven by

$$\text{future congestion risk}$$

rather than

$$\text{current congestion.}$$

However, if the prediction interval for e_1 were extremely wide because twin fidelity was poor, the controller could retain the existing path because the expected advantage would no longer compensate for prediction risk and route-reconfiguration cost.

This behavior is the central idea of the proposed framework.

N. Model-Predictive Operation

The proposed system operates using a receding-horizon mechanism.

At epoch t:

$$\text{Telemetry} \rightarrow \text{Twin synchronization} \rightarrow \text{Fidelity estimation}$$

$$\rightarrow \text{Future-state scenarios} \rightarrow \text{Counterfactual simulation} \rightarrow \mathbf{P}1.$$

The optimizer produces

$$\{x^*_{fph} : h = 1, \ldots, H\}.$$

The controller installs only the first action

$$x^*_{fp1}$$

in the physical SDN.

At epoch $t + 1$, new telemetry arrives, the twin is resynchronized, prediction errors are measured, uncertainty bounds are recalibrated, and the optimization problem is solved again.

Hence,

$$\boxed{\text{Predict} \rightarrow \text{Simulate} \rightarrow \text{Risk-evaluate} \rightarrow \text{Act} \rightarrow \text{Observe} \rightarrow \text{Recalibrate.}}$$

This closed-loop design prevents long-horizon forecast errors from permanently controlling the physical network.